

Payment instructions are settled under the CLS Bank rules when it debits and credits the relevant Settlement members' accounts for the amounts involved. This final settlement will only occur within the context of several risk management tests. The risk management tests are: (1) any Short Position in respect to each currency would not exceed the relevant Settlement Member's Short Position Limit (adjusted based on the committed liquidity facilities available to CLS Bank for that day); (2) the Aggregate Short Position in the Settlement Member's Account, calculated as an equivalent U.S. dollar value, would not exceed the Aggregate Short Position Limit which is set for that Settlement Member; and (3) the sum of the Settlement Member's Long Positions and Short Positions (the "Account Balance"), adjusted by applicable Haircuts (the "Adjusted Account Balance"), would not be negative. If any of these tests are not satisfied, no debits or credits will be made, and the pair of Settlement Eligible Instructions will remain on the settlement processing queue and be retested each time the queue is cycled through. The risk management tests will continue to be applied to all Settlement Eligible Instructions on the settlement processing queue until the applicable currency closing time has passed or all Settlement Eligible Instructions on the settlement processing queue have been settled. CLS Bank does not (i) guarantee the settlement of every payment instruction that is submitted for settlement or (ii) become a counterparty to any FX transaction referenced in a payment instruction that is submitted for settlement. In the event that a member is unable to provide its expected pay-in, other CLS members may be obligated to provide revised pay-in requirements.

## **Securities Settlement Systems**

The major securities markets in the United States include the government securities market, the corporate equities and bond market, the market for money market instruments, and the municipal bond market. The instruments traded in these markets are generally traded through organized exchanges or through the over-the-counter dealer markets. Major categories of financial instruments commonly traded include U.S. Treasury securities, government agency securities, securities issued by federal government-sponsored enterprises, corporate equities and bonds, money market instruments such as commercial paper, and municipal (state and local) government securities. Participants in these markets include securities issuers; intermediaries such as brokers and dealers; and investors such as insurance companies, investment companies, non-financial corporations and individuals.

The mechanisms for clearing and settling securities transactions vary by market and type of instrument, and generally involve two types of specialized financial intermediaries: clearing corporations and securities depositories. Clearing corporations provide trade confirmation and comparison services, and multilateral netting of trade obligations; while, securities depositories transfer securities ownership on a gross or net basis against payment via book entry transfers.

Depository institutions play several important roles in securities clearing and settlement. Not only do depository institutions participate in clearing and settlement arrangements for themselves, they also act as custodians, issuing and paying agents, and settling banks for their customers. In addition depository institutions provide credit to clearing corporations, securities depositories, and clearing participants for routine and contingency purposes.

## U.S. Government Securities

The U.S. government securities market encompasses all primary and secondary market transactions in securities issued by the U.S. Treasury, certain federal government agencies, and federal government-sponsored enterprises.<sup>[8]</sup> Trading in government securities is conducted over the counter between brokers, dealers, and investors. In over-the-counter trading, participants trade with one another on a bilateral basis rather than on an organized exchange. Nearly all U.S. government securities are issued and transferred through a book-entry system operated by the Federal Reserve.

In the primary market, U.S. Treasury securities are issued through regularly scheduled auctions. The Federal Reserve Banks serve as conduits for the auctions, with the Federal Reserve Bank of New York coordinating much of the auction activity. Individuals, corporations and financial institutions may participate in the auctions. Participation in Treasury auctions, however, is typically concentrated among a small number of dealer firms, known as primary dealers.<sup>[9]</sup>

In the secondary market for government securities, trading activity takes place between primary dealers and non-primary dealers. Customers of these dealers are financial institutions, non-financial institutions and individuals. The majority of transactions between primary dealers and other large market participants are conducted through inter-dealer brokers that provide both anonymity and price information to market participants. Approximately 2,000 securities brokers and dealers are registered to operate in the U.S. government securities market.

### Fixed Income Clearing Corporation (FICC)

FICC, FICC is a subsidiary of the Depository Trust and Clearing Corporation, a holding company that also includes the National Securities Clearing Corporation and the Depository Trust Company. In January 2003, the Mortgage-Backed Securities Clearing Corporation merged with the Government Securities Clearing Corporation to form the FICC. For further information about the Government Securities Division and the Mortgage-Backed Securities Division, see [www.ficc.com](http://www.ficc.com). composed of the Government Securities Division (GSD) and the Mortgage-Backed Securities Division (MBSD), compares and nets trades of U.S. Treasury securities, agency debt securities, and mortgage-backed securities. As the name implies, GSD clears and nets U.S. government securities and agency debt securities. MBSD provides automated post-trade comparison, netting, risk-management, and pool notification services to the mortgage-backed securities market. Securities eligible for MBSD clearing are mortgage-backed securities issued by the Government National Mortgage Association (GNMA), the Federal Home Loan Mortgage Corporation (FHLMC), and the Federal National Mortgage Association (FNMA).

FICC uses real time trade matching; trade details are compared and matched as soon as

trade information is submitted. Successfully compared trades result in binding and enforceable obligations for settlement. Unmatched trades may be revised to achieve a trade match.

Successfully matched trades of eligible securities, for FICC netting service participants, are netted against offsetting net-receive or net-deliver obligation of the same security. Once the government securities net positions are determined GSD interposes itself between the original trading parties and becomes the legal counter-party to FICC members for settlement purposes. Therefore, GSD members' net settlement obligations are delivered to or received from GSD. MBSD, however, engages in multilateral position netting and does not stand in the middle of transactions. Final net settlement obligations of GSD and MBSD participants are settled through the Fedwire Securities Service via participants' settlement bank.

## **Fedwire Securities Service**

As fiscal agents of the United States, the Federal Reserve Banks act as the securities depository for all marketable U.S. Treasury securities, many federal agency securities, and certain mortgage-backed securities issued by GSEs. The Federal Reserve also acts as agent and depository for the securities of certain international organizations, such as the World Bank. U.S. government securities are issued in book-entry form through the Federal Reserve's Fedwire Securities Service using either an auction process or dealer syndicate mechanisms. The Federal Reserve's Fedwire Securities Service provides for the safekeeping and transfer of these securities. The safekeeping function involves the records of securities balances, and the transfer and settlement function involves the transfer of securities between parties.

When book-entry securities transfers are processed using Fedwire Security Service, the institution sending the transfer receives immediate credit in its Federal Reserve (funds) account for the payment associated with the transfer, and its securities account is correspondingly debited. The Federal Reserve (funds) account of the institution receiving a book-entry securities transfer is debited for the payment amount, and its securities account is credited. There are more than 9,000 participants in the system and they are largely composed of depository institutions.

The Federal Reserve's Fedwire Securities Service is supported by a real-time, delivery-versus-payment (DVP) gross settlement system that provides for the immediate, final, and simultaneous transfer of securities against the settlement of funds. This system, known as the National Book-Entry System (NBES), provides for the safekeeping and transfer of U.S. Treasury, government agency, and GSE securities as well as securities issued by certain international organizations. The safekeeping function involves the electronic storage of securities records in custody accounts, and the transfer and settlement function involves the transfer of securities between parties.

Financial institutions may access the Fedwire Securities Service via high-speed direct CI, FedLine, or with off-line telephone connectivity with a Federal Reserve Bank. Financial institutions may also access certain Fedwire Securities Service inquiry information via FedLine for the Web. On-line participants, using either a mainframe or FedLine PC connection to Fedwire Securities Service, require no manual processing by the Federal Reserve Banks. Off-line participants provide funds transfer instructions to their Federal Reserve Bank by telephone, and once authenticated, the Federal Reserve Bank enters the transfer instruction into the Fedwire Securities Service system for execution. The manual processing required for off-line requests makes them more costly

and suitable only for institutions processing a small number of funds transfer payment orders.

## **Corporate and Municipal Securities**

Corporate equities and bonds, commercial paper, and municipal bonds are traded on various established exchanges and on over-the-counter markets. The primary securities exchanges in the United States are the New York Stock Exchange, the American Stock Exchange, and the primary over-the-counter dealer market is the National Association of Securities Dealers Automated Quotations (NASDAQ).

The commercial paper market warrants special mention as it is an important source of short term funding for financial corporations and municipal governments. Commercial paper is a money market instrument issued by prime-rated non-financial and financial companies with maturities ranging from one to 270 days. Commercial paper is issued through dealer placements or direct placements with investors. Although commercial paper is a negotiable instrument, secondary market trading is limited. Disruption in the issuance of commercial paper can cause significant liquidity and credit concerns for issuers, many of whom are depository institutions and other financial companies, and depository institutions that provide issuing and paying agent services to issuers.

### **National Securities Clearing Corporation (NSCC)**

NSCC, established in 1976, provides clearing and settlement services for corporate equities, corporate debt, municipal securities, mutual funds, annuities, and unit investment trusts. NSCC is a registered clearing corporation regulated by the Securities and Exchange Commission (SEC). With roughly 250 full-service participants, NSCC handles all aspects of the clearing and settlement of trades between brokers and dealers in securities traded on the over-the-counter markets, the New York Stock Exchange, the American Stock Exchange, and other regional exchanges.

Executed trades are typically reported to NSCC on trade date. Trades are either reported to NSCC by an established exchange as matched (irrevocable) trades or by brokers and dealers as unmatched trades. Trades submitted by brokers and dealers are compared and matched within NSCC's Continuous Net Settlement (CNS) system. Following the comparison process, NSCC becomes the legal central counterparty and guarantees completion of all securities trades following through CNS. CNS functions as an automated book-entry accounting system that centralizes the settlement of compared security transactions and maintains an orderly flow of securities and money balances. Further information about NSCC's clearing processes can be found at [www.nscc.com](http://www.nscc.com).

Similar to the FICC netting process, successfully matched trades of eligible securities are netted against offsetting net-receive or net-deliver obligation of the same security.

Settlement of securities takes place at the Depository Trust Company and funds settlement occur over the Fedwire Funds Service through settlement banks.

### **Depository Trust Company (DTC)**

DTC is a New York limited purpose trust company, a member of the Federal Reserve System, and a clearing agency registered with the SEC. DTC, the central securities depository for corporate equities and bonds, municipal government securities, and money market instruments, provides safekeeping and transfer of these securities. DTC participants include securities brokers, dealers, institutional investors, and depository institutions acting on their own behalf as well as functioning as custodians, issuing and paying agents, and settling banks for their customers.

Like Fedwire Securities, the safekeeping function involves the electronic recordkeeping of securities balances for participants, and the transfer function involves the transfer of securities between parties. DTC transfers securities on a gross basis throughout the day, while settling funds obligations on a net basis at the end of the day. During the transfer process, DTC will limit its credit exposure through participants' net debit caps. Any net debit position incurred by participants must be fully collateralized. End of day funds settlement occur over the Fedwire Funds Service through participants' settlement banks. See [www.dtc.org](http://www.dtc.org) for more information on DTC.

## **Intrabank Payment and Messaging Systems**

### **Internally Developed and Off-The-Shelf Funds Transfer Systems**

#### ***Action Summary***

Financial institutions require efficient systems for transferring funds internally, among themselves, and with their customers for large-dollar payments relating to financial market transactions and settling corporate and consumer payments.

Management and the board should:

- Establish dual controls and separation of duties for funds transfer systems;
- Monitor and log access to funds transfer systems, maintaining an audit trail of all sequential transactions; and
- Incorporate the funds transfer controls into the organization's information security program to ensure the integrity and confidentiality of customer information.

Financial institutions rely on internal funds transfer systems and networks to send payment instructions to their correspondents for the transfer of correspondent balances or to initiate Fedwire Funds Service or CHIPS payments. Large financial institutions have either developed their own funds transfer systems or relied on off-the-shelf funds transfer systems. In either case, the internal financial institution funds transfer systems interface with Fedwire Funds Service and CHIPS, supporting the interface and transaction format specifications for the transmission of payment orders. Off-the-shelf funds transfer systems typically support a variety of treasury, cash management, and straight-through-processing (STP) modules, which automate payment order processing.

The Federal Reserve Banks provide the Computer Interface Protocol Specifications (CIPS) that funds transfer and book-entry securities systems need to adopt in order to implement a CI connection successfully. The Federal Reserve provides a website with a list of vendors who have completed the Federal Reserve Banks' protocol certification process.<sup>[10]</sup> The Federal Reserve Banks do not endorse any specific software vendor or product. The Federal Reserve Banks make no warranties or representations with respect to any of the products offered by these vendors except that communication-level software correctly executes systems network architecture (SNA) commands as specified in the CIPS.

## **Payment Messaging Systems**

Financial institutions, corporations, and other organizations employ wholesale payment message systems to originate payment orders, either for their own benefit or for a third party. These systems are indispensable components of funds transfer activities. Unlike payment systems, which transmit actual debit and credit entries, message systems process administrative messages and instructions to move funds. The actual movement of the funds is then accomplished by initiating the actual entries to debit the originating customer's account and credit the beneficiary's account at one or more financial institutions. If the beneficiary's account or the beneficiary institution's account is also with the originator's institution, the institution normally handles the transaction internally through a book transfer. If the beneficiary related accounts are outside the originating customer's institution, the parties will complete the transfer by use of a payments system such as Fedwire Funds Service or CHIPS. The means of arranging payment orders range from manual methods (e.g., memos, letters, telephone, fax, or standing instruction) to electronic methods using telecommunications networks. These networks may include those operated by the private sector, such as SWIFT or Telex, or operated internally by or for the institution. The internal networks can be for inter-company purposes only or connected to customer sites.

Since the payment order is the institution's authorization to act on behalf of the customer, it is imperative that a system is in place to establish the authenticity and time of receipt of the order. These two elements are the primary components cited by the Uniform Commercial Code Article 4A (UCC4A) in establishing responsibility for the execution of a payment order. Even though the transfers initiated through systems such as SWIFT and

Telex do not result in the immediate transfer of funds from the issuing institution, they do result in the issuing institution having an immediate liability, which is payable to the disbursing institution. Therefore, the physical and logical controls surrounding payments messaging systems should include:

- Physical controls limiting access to only those staff members assigned responsibility for managing the payment messaging system;
- Logical access controls restricting access on a need to know basis;
- Assigning access to payment messaging application and data based on functional job duties and requirements; and
- Identification and authentication controls used to authenticate access to payment messaging systems.

### **In-house Terminals**

Some financial institutions employ terminals, connected via telecommunications networks with customers and the institutions' operating departments, to execute funds payment orders. These systems may be dial-up or dedicated lines and are often fully interfaced to the institution's funds payments system. The primary security method is the use of unique passwords for each user of the system. Since there is often no intervention by the funds payment operation, it is necessary to establish controls directly in the area employing the terminals. These controls should cover origination, data entry, and release, and should be the same as those associated with an independent funds payment function.

### **Non-automated Payment Order Origination**

While in-house terminals are the primary sources for payment order origination, less complex institutions still rely heavily on memos, letters, telephone, fax, or standing instructions. (Note: standing instructions are normally maintained in the automated funds transfer system as recurring transfers and should be subject to the same input/verification controls as wires when first entered into the system). It is imperative that an institution using these payment order methods has a viable security program, which includes:

- Maintaining signature lists for use with internally and externally generated memos, letters, or fax instructions. As noted in UCC4A Section 201, signature verification alone is not defined as a security procedure; however, institutions may use it with other security devices such as call backs or codes.
- Call back to authorized individuals for both internally and externally generated telephone instructions.
- Procedures covering standing instructions protecting against unauthorized change, periodic review to validate accuracy, and ensuring execution under the agreed

terms.

## **Funds Transfer Operations (Wire Room)**

A financial institution's funds transfer operation (wire room) is responsible for originating, transmitting, and receiving payment orders. In less complex financial institutions, the wire room typically includes a FedLine PC.<sup>[11]</sup> Less complex institutions may also have a core banking package that includes a funds transfer module, which generates payment orders in a Fedwire Funds Service format for uploading to the FedLine PC. Staff assigned responsibility for these activities are generally responsible for other duties and are not typically dedicated full-time to the wire room function. In most financial institutions, funds transfer payment order volume does not justify the costs associated with a full time staff, and the sending and receiving of payment orders may be a part-time responsibility for one or more people. For less complex financial institutions, a complete separation of duties may be difficult to achieve, and compensating controls, including rotation of duties and internal review procedures covering those payment orders requiring officer review, should be considered.

Financial institutions generating significant payment order volume usually have a separate funds transfer department with dedicated staff. Financial institutions generating a large volume of high value Fedwire Funds Service payment orders typically use dedicated funds transfer software (developed in-house or purchased) connected via computer interface to the Federal Reserve Bank's Fedwire Funds Service application. The software used for wire transfers automatically posts transactions to the demand deposit account and general ledger. The automated function provides an efficient means to process a large number of payment orders supporting a variety of business lines.

Payment orders can be received from several different sources including business areas within the financial institution, as well as from corporate and individual customers. Payment orders can be initiated by phone, fax, and online systems. Individuals wishing to wire funds typically do so at the teller window or contact their loan officer or account representative. Payment order verification is an important safeguard, and institutions should, at a minimum, keep accurate records of all payment order requests, including those initiated by telephone. Institutions should record all phone calls initiating payment orders for security and audit reasons. The institutions should maintain the tapes for at least a 30-day period.

After receiving a payment order, the wire room operator keys the payment order into FedLine (or the payment order is generated through the use of a third-party software product funds transfer module). Before sending a payment order to the Federal Reserve Bank, a second staff member should verify it for accuracy and authorization. Most FedLine PCs have two printers attached, one that prints copies of all outgoing payment order Fedwire Funds Service messages and another that prints incoming Fedwire Funds Service payment order messages. Institutions should maintain a record of all payment orders for record keeping purposes. The unbroken printout sheet helps ensure a complete record of all messages; however, institutions should also verify the sequence numbers of the messages to identify missing records due to communication problems. The sequence number provides an audit trail for all funds transfers on the Fedwire Funds Service system.

The institution should have appropriate procedures in place to verify all processed payment orders. These procedures usually include the use of code words, call backs,



and corporate resolutions authorizing certain employees to send payment orders. Verification and security procedures are extremely important in light of the potential for fraud or errors.

A Fedwire Funds Service message is generated either by the application supporting the business line or by an authorized wire room employee who enters the message into an on-line terminal. Before transmitting the wire, it is sent to a second terminal for an independent employee to verify for accuracy as well as proper authorization. Only after a second staff member reviews the payment order should a financial institution send it to the Federal Reserve Bank for processing.

This separation of duties is important to ensure security. The institution's internal funds transfer system should maintain data on each day's transfers, including wires sent and received, wires listed by amount, wires listed by sequence number, and wires listed by account holder. Most software systems maintain the work of several previous days, often the last 5 to 7 days, to allow on-line access to trace errors and problems. After the 5 to 7 days, the data is typically archived.

## **Computer and Network Operations Supporting Funds Transfer**

Wholesale funds transfer systems are high risk. Therefore, management should configure hardware and software components to control access and support effective monitoring. Management should develop change management procedures to ensure the integrity of the hardware configurations and applications software. Operations personnel should have the appropriate procedures to manage critical payment systems software.

Applications should employ strong user authentication, support user entitlement (information access and function controls) administration, and provide audit trails in sufficient detail to support the analysis or investigation of specific transactions. Management should enable funds transfer activity logs and designate independent staff members to monitor operations, applications support, system administration, and security administrators' activities associated with the funds transfer system.

Telecommunications systems employed for EFT can range from a dial-up connection between the institution and payments system (e.g., FedLine) to terminal connections with institution staff and customers that transmit institution's funds transfer system payment orders directly to Fedwire Funds Service via CI connection. An institution's information security program should include access, authentication, and transmission controls surrounding wire room activities and all terminal connections. Access and authentication controls may consist of personal identification numbers, passwords, or other identifying keys such as account numbers, balances, or other financial data. Financial institutions should use encryption as a means of protecting data throughout the EFT system. Encrypting data during transmission allows institutions to scramble the contents of message/payment orders during transmission and limit the value of the information to an interloper even if a transmission is intercepted. Nevertheless, financial institutions should monitor or prevent access to funds transfer activity by data processing personnel who

have access to communications equipment and can monitor and record data flowing in clear text from encryption devices.

# Wholesale Payment Systems Risk Management

## Payments System Risk (PSR) Policy

Given the role of payments and securities settlement systems as critical components of the nation's financial system, the Federal Reserve Board has developed a policy to foster the safety and efficiency of these systems. The regulatory agencies have also issued interagency and agency-specific policies on payments and payment system risk. The "Interagency Paper on Sound Practices to Strengthen the Resilience of the U.S. Financial System," for example, identifies sound practices focusing on minimizing the immediate systemic effects of a wide-scale disruption to critical financial markets. The sound practices discussed in this paper supplement the agencies' respective policies and other guidance on business continuity planning. See [http://www.federalreserve.gov/paymentsystems/psr\\_policy.htm](http://www.federalreserve.gov/paymentsystems/psr_policy.htm)

One major component of the Federal Reserve's policy addresses the extent to which institutions with Federal Reserve accounts are allowed to have daylight overdrafts in that account. Additional information about this aspect of the PSR policy is provided in Appendix E. A daylight overdraft occurs when an institution's Federal Reserve account is in a negative position during the business day. Negative balances typically occur when there are insufficient funds in an institution's account to cover outgoing Fedwire funds transfers, incoming book-entry securities transfers, or other payment activity processed by a Federal Reserve Bank. The PSR policy applies a series of limits and incentives to control the Federal Reserve Banks' exposures to the credit risk associated with these daylight overdrafts. Limits on net debit positions are sufficiently flexible to reflect the overall financial condition and operational capacity of each institution using Federal Reserve Bank payment services.

Institutions with the highest net debit caps are subject to an annual self-assessment process that is outlined in the "Guide to the Federal Reserve's Payments System Risk Policy." [http://www.federalreserve.gov/paymentsystems/psr\\_policy.htm](http://www.federalreserve.gov/paymentsystems/psr_policy.htm) In addition to an analysis of the institution's general creditworthiness, institutions eligible for a self-assessed cap should manage and control their intraday funds positions, maintain credit policies and controls at the customer level, and employ operational controls and contingency plans. In particular, institutions with self assessed caps need to be able to monitor and control the intraday credit they extend to their customers. Such credit has the potential to result in Federal Reserve daylight overdrafts for the institution. The risks associated with customers' daylight overdrafts are discussed below.

Another major component of the PSR policy provides expectations for risk management in payments and securities settlement systems, and adopts international standards for systemically important systems. See [http://www.federalreserve.gov/paymentsystems/psr\\_policy.htm](http://www.federalreserve.gov/paymentsystems/psr_policy.htm) for details and further references. This portion of the policy is directed at system operators, but contains key information about the risks that such systems present to system participants, and in some cases, requires covered systems to provide participants with clear information about these risks. Moreover, examined institutions may have an important role in the governance of wholesale payments and securities settlement systems, as many private sector systems are owned by their participant